Some Important Issues: Multicollinearity and Outliers

November 14, 2022

Looking Ahead:

1. Today: Important Issues in Multiple Regression: Multicollinearity and Outliers
2. Thursday (November 10): Logistic Regression
3. Tuesday (November 15): In-class Regression Lab- Due November 29 if not completed
   - **Checkpoint 3 Due 11:59PM: Preliminary Analysis**
4. Thursday (November 17): No class!

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Please come see me <u>THIS WEEK</u> if you need help! By appt. or in OH

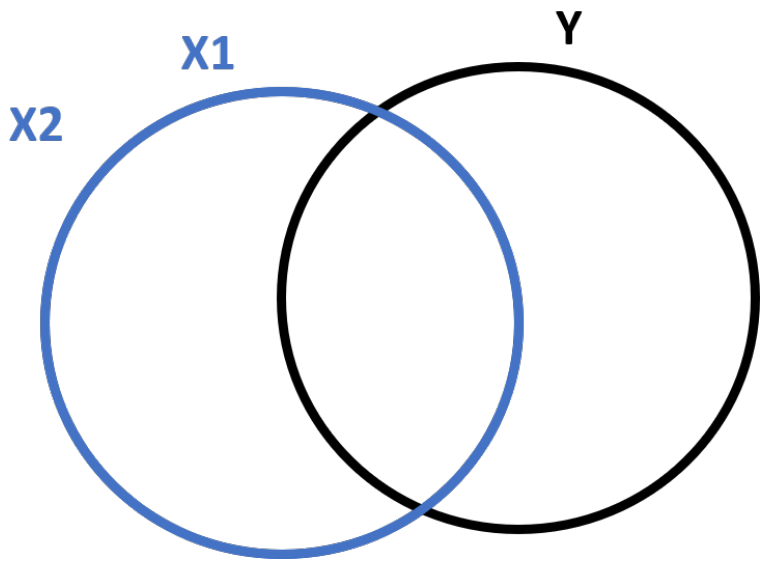
Multicollinearity

Estimating multiple regression requires one more assumption (compared to bivariate regression):

Multicollinearity

Estimating multiple regression requires one more assumption (compared to bivariate regression): **There is no perfect multicollinearity.**

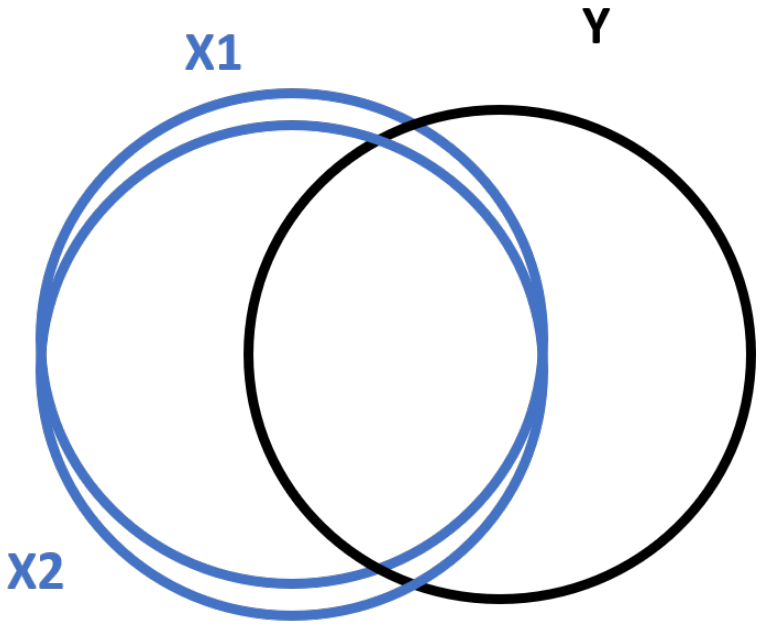
- *Perfect Multicollinearity*: When one independent variable is an exact linear function of one or more other independent variables in a model.



In practice, pretty rare.

Multicollinearity

High Multicollinearity: When one independent variable is highly correlated with one or more other independent variables in a model.



This can happen a lot in practice. Consequences?

- Hard to discover statistically significant relationships
- Hard to determine the precise effects of each individual variable

Multicollinearity: How do I Know?

1. Quick Check: High R^2 but few statistically significant variables
2. Correlation Matrix
3. Variance Inflation Factor (VIF) diagnostic

Multicollinearity Example

Who voted for Clinton's Impeachment in US HR?

- `pctbl96` – the percentage of that member's House district that was African-American, as of 1996.
- `unionpct` – the percentage of that member's House district that were members of organized labor,
- `clint96` – the percentage of the two-party presidential vote that President Clinton received in that district in the 1996 election,
- `GOPmember` – a dummy variable coded 1 if the representative was a member of the Republican party, and 0 if they were a Democrat, and
- `ADA98` – the member's 1998 Americans for Democratic Action score; a 0-100 measure of how liberal that member's voting behavior is.

Multicollinearity Example

```
> fit<-lm(votesum~ADA98+GOPmember+clint96+pctbl96+unionpct)
> summary(fit)

Call:
lm(formula = votesum ~ ADA98 + GOPmember + clint96 + pctbl96 +
    unionpct)

Residuals:
   Min     1Q Median     3Q    Max
-3.271 -0.259  0.133  0.337  2.731

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.51785    0.23246   10.83   <2e-16 ***
ADA98       -0.02144    0.00238   -9.00   <2e-16 ***
GOPmember    1.59981    0.18043    8.87   <2e-16 ***
clint96     -0.00935    0.00433   -2.16    0.031 *
pctbl96      0.00347    0.00270    1.29    0.199
unionpct    -0.52544    0.48065   -1.09    0.275
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

Residual standard error: 0.629 on 427 degrees of freedom
Multiple R-Squared: 0.883,Adjusted R-squared: 0.882
F-statistic:  647 on 5 and 427 DF,  p-value: <2e-16
```

Correlation Matrix

```
> cor(idata)
          district  votesum  pctbl96 unionpct clint96 GOPmember    ADA98
district   1.00000 -0.03496 -0.06759  0.09155  0.1044  -0.02881  0.04988
votesum   -0.03496  1.00000 -0.28765 -0.26199 -0.6408   0.91977 -0.92795
pctbl96   -0.06759 -0.28765  1.00000 -0.09394  0.6165  -0.30911  0.30288
unionpct   0.09155 -0.26199 -0.09394  1.00000  0.3331  -0.19406  0.27563
clint96    0.10437 -0.64084  0.61651  0.33305  1.0000  -0.61196  0.67033
GOPmember -0.02881  0.91977 -0.30911 -0.19406 -0.6120   1.00000 -0.93918
ADA98      0.04988 -0.92795  0.30288  0.27563  0.6703  -0.93918  1.00000
```

Variance Inflation Factor (VIF)

For each variable X_j in model:

1. Run a regression with X_j as the dependent variable and the other independent variables as the model
2. Take R_j^2 from each model and plug into equation:

$$VIF_j = \frac{1}{(1 - R_j^2)} \tag{1}$$

3. Higher VIF score means X_j may be highly correlated with one or more other independent variables
   - Above 2.5 may be concerning
   - Above 10 means very high multicollinearity

Variance Inflation Factor (VIF) Code

```
> vif(fit)
    ADA98 GOPmember   clint96   pctbl96  unionpct
   10.292     8.878     3.313     1.998     1.371
```

High Multicollinearity: What To Do?

Multicollinearity is a *data* problem, not a statistical one.

- Ideal: Add more data
- Should we drop one of the violators?
  - THEORY!
  - Remember the dangers of omitted variable bias!
- Live with it and be transparent

Outliers

Outliers

Linear regression, like the sample mean, is not robust to outliers

In regression contexts, **outliers** are *disproportionately influential* observations.

- *leverage*- observation with unusual values on one or more independent variables

Consequences?

Detecting Outliers

1. Plot your data!!!
2. DFBETAS
3. Tons of others, like Cook's D, DFFITS, etc.

Detecting Outliers

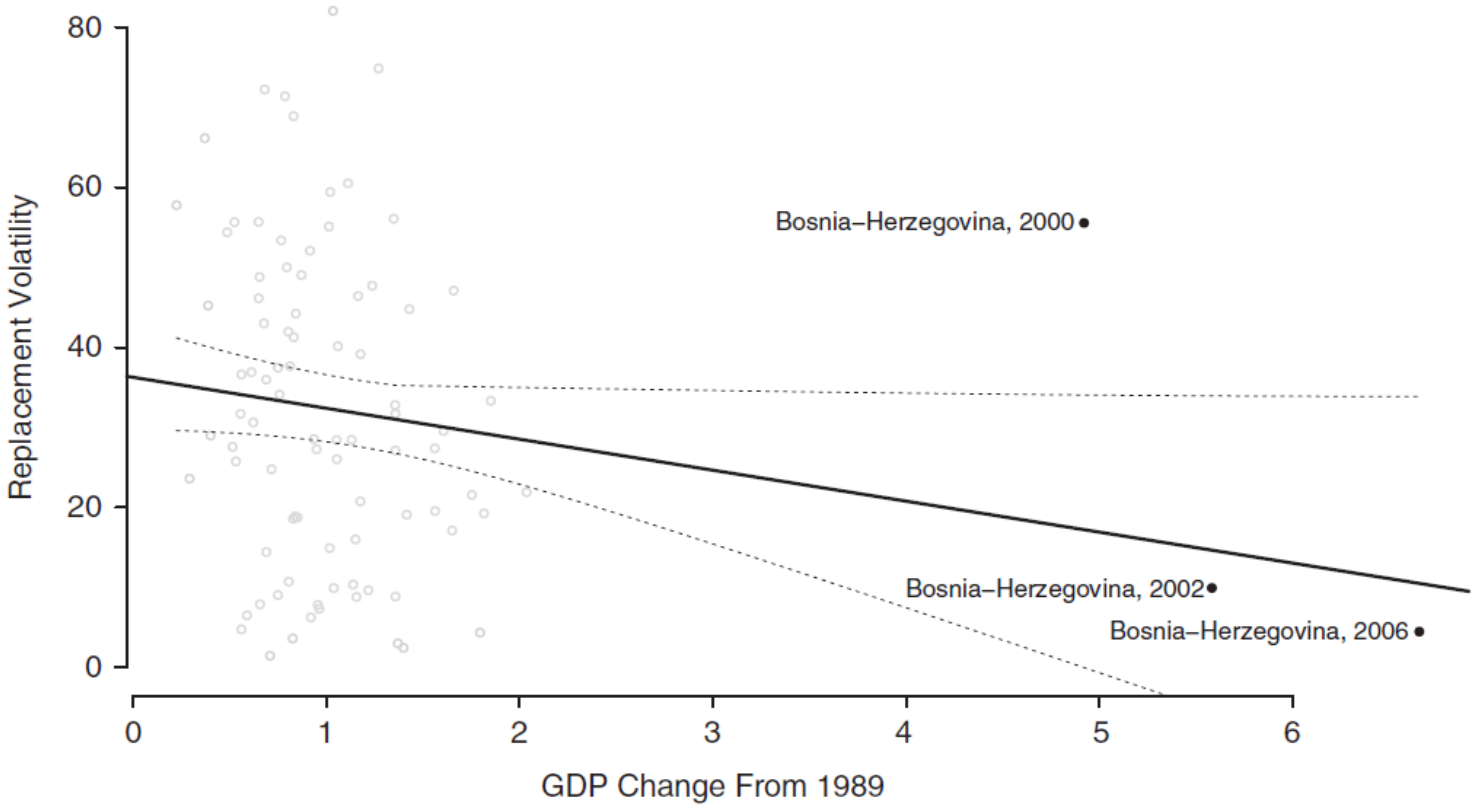
Claim: Replacement Volatility (party volatility caused by new party entry and old party exit) is driven by long-term economic performance.

Results

TABLE 1 *Electoral and Replacement Volatility in Post-Communist Europe*

	Verification	
	Electoral Volatility	Replacement Volatility
GDP Change from 1989	0.639	–4.623***
	(0.693)	(1.326)
GDP Change Between Elections	–2.059	9.019
	(5.219)	(10.128)
Effective Number of Electoral Parties	0.446	–0.346
	(0.313)	(0.533)
Log-Weighted District Magnitude	–0.784	0.638
	(0.887)	(2.931)
Presidential System	–4.631	6.784
	(4.126)	(9.435)
Semi-presidential System	–2.788	4.255
	(2.211)	(5.897)
Proportional Representation	0.827	0.077
	(2.228)	(6.004)
Ethnic Fractionalization	–6.163	–2.677
	(6.397)	(18.978)
Years Since Collapse of Communism	0.848	–2.633
	(0.807)	(2.153)
Years Since Collapse Squared	–0.031	0.070
	(0.042)	(0.101)
Constant	13.059**	41.941***
	(5.318)	(13.329)
Countries	21	21
Pairs of Elections	89	89
R^2	0.116	0.139

Plot


Replacement Volatility
80
60
40
20
0
Bosnia–Herzegovina, 2000
Bosnia–Herzegovina, 2002
Bosnia–Herzegovina, 2006
0
1
2
3
4
5
6
GDP Change From 1989

Results

TABLE 1 *Electoral and Replacement Volatility in Post-Communist Europe*

	Verification		Without Bosnia-Herzegovina	
	Electoral Volatility	Replacement Volatility	Electoral Volatility	Replacement Volatility
GDP Change from 1989	0.639	−4.623***	0.116	−6.066
	(0.693)	(1.326)	(3.206)	(7.178)
GDP Change Between Elections	−2.059	9.019	−1.891	6.677
	(5.219)	(10.128)	(5.898)	(10.704)
Effective Number of Electoral Parties	0.446	−0.346	0.452	−0.264
	(0.313)	(0.533)	(0.316)	(0.558)
Log-Weighted District Magnitude	−0.784	0.638	−0.789	0.603
	(0.887)	(2.931)	(0.882)	(2.893)
Presidential System	−4.631	6.784	−4.847	5.532
	(4.126)	(9.435)	(4.606)	(10.241)
Semi-presidential System	−2.788	4.255	−2.813	4.017
	(2.211)	(5.897)	(2.286)	(5.885)
Proportional Representation	0.827	0.077	0.852	−0.146
	(2.228)	(6.004)	(2.265)	(5.943)
Ethnic Fractionalization	−6.163	−2.677	−6.716	−5.298
	(6.397)	(18.978)	(6.784)	(22.939)
Years Since Collapse of Communism	0.848	−2.633	0.828	−1.989
	(0.807)	(2.153)	(0.863)	(2.117)
Years Since Collapse Squared	−0.031	0.070	−0.029	0.045
	(0.042)	(0.101)	(0.044)	(0.097)
Constant	13.059**	41.941***	13.586***	43.661***
	(5.318)	(13.329)	(5.115)	(14.057)
Countries	21	21	20	20
Pairs of Elections	89	89	86	86
R^2	0.116	0.139	0.114	0.119

DFBETAS

Calculated as the difference in each $\hat{\beta}$s with and without each individual observation, divided by the original standard error estimate of the given $\hat{\beta}$.

- So, bigger differences mean observation is more influential, because β changes more when it is excluded
- In practice, DFBETAS cutoff for "unusually influential" observations is $\frac{2}{\sqrt{n}}$

Code:

- dfbetas(model)

Interpretation:

- Look at all observations (rows) for all variables (columns) and see if any are bigger than $\frac{2}{\sqrt{n}}$

Dealing with Outliers

- Double-check values to make sure not a mistake
- Transform variable (e.g., logarithm if consistent with a skew pattern)
- "Dummy out" influential case: make a binary variable for that observation only and include in model
  - E.g., create variable for U.S. (1 if country is U.S., 0 otherwise), then include this U.S. variable in model
- Only drop observations in extreme circumstances